

Hybrid Retrieval with Reciprocal Rank Fusion

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Abstract. We study how dense and sparse retrievers can be combined. Reciprocal Rank Fusion (RRF) merges ranked lists without score calibration and improves nDCG@10 by 4.1 points on our benchmark.

1 Method

Given ranked lists from m retrievers, the fused score of document d is defined in Eq. (1):

$$s(d) = \sum_{i=1}^m \frac{w_i}{k + r_i(d)} \quad (1)$$

BM25 scores a term t in document d with the saturation function in Eq. (2):

$$\text{bm25}(t, d) = \text{idf}(t) \cdot \frac{f(t, d) (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}}\right)} \quad (2)$$

2 Results

Table 1: Retrieval quality on the test set.

Method	Recall@10	MRR@10	nDCG@10
BM25	0.712	0.534	0.581
Dense	0.768	0.571	0.619
Hybrid (RRF)	0.823	0.602	0.660

Algorithm 1 Reciprocal Rank Fusion

```
Input: ranked lists L1..Lm, weights w, constant k
1: for each document d in union(L1..Lm) do
2:   s(d) <- sum_i w_i / (k + rank_i(d))
3: end for
4: return documents sorted by s(d) descending
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References

- [1] G. Cormack, C. Clarke, S. Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. SIGIR 2009.
- [2] S. Robertson, H. Zaragoza. The probabilistic relevance framework: BM25 and beyond. 2009.